PROGRAM SUMMARY

Program title: readcon-core
CPC Library link to program files: (to be added by Technical Editor)
Developer's repository link: https://github.com/lode-org/readcon-core
Licensing provisions: MIT
Programming language: Rust, C, C++, Python, Julia, Fortran 2008
Supplementary material: specification, grammar, conformance corpus, and the tables behind the measurements

Nature of problem: Rare-event campaigns stop and resume from a stored geometry. The formats in common use each drop part of the resume state: a per-axis mask, a persistent atom identifier, the velocity time unit, binary64 coordinates, or the periodic flags. Readers of one format also disagree with each other, so the resumed search runs on a different surface.

Solution method: readcon-core 0.17.1 implements CON 3.1, a versioned grammar. A reader of a known major accepts a later minor, keeps unknown keys, and rejects an unknown major. Every velocity block declares its time unit. The default writer emits the shortest decimal that parses back to the same binary64 value. A corpus of legal and illegal frames decides what a conforming reader accepts. One Rust core sits behind one C interface, and eOn 3.6.0 reads and writes CON through it. The companion store keeps that text, indexes composition, energy and forces, recognises gzip and zstd wrappers of the same text, and ingests frames from several MPI ranks. Its map starts at 2 GiB, and the caller can raise that limit.

Additional comments including restrictions and unusual features: A frame stores geometry and its labels. The short history of a quasi-Newton or FIRE step lies outside the frame.

CON 3.1: a versioned restart state for saddle searches

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Abstract

A saddle search resumes from a geometry file, so that file is the restart state of the algorithm. The resume needs a per-axis mask, a persistent atom identifier, velocities in a declared time unit, coordinates that round-trip in binary64, and a versioned grammar. On one masked Pt(100) frame, every common format tested here drops at least one of those fields. An ASE POSCAR round trip drops the adatom's height constraint, and the resumed climbing-image band finishes 70.2 meV below the uninterrupted barrier of 0.644561366 eV. Five extended XYZ readers return four answers for one file. CON 3.1 is the versioned grammar of the text eOn already writes, checked by a conformance corpus and implemented once: a Rust core behind a C interface for C, C++, Python, Julia and Fortran. Twenty-five language pairs resume that band at the uninterrupted barrier. A warm read of 10,000 frames of 218 atoms, recorded with readcon-core 0.17.0, takes 12.6 μs per frame. On that file chemfiles is 2.15 times as fast and h5py is 67.1 times as fast. The release reported here is readcon-core 0.17.1.

Keywords: restart, specification, saddle search, conformance, nudged elastic band

1. Introduction

Rare-event methods turn a potential energy surface into rates. Transition state theory [1, 2, 3] reduces kinetics to saddle points. These are located by the nudged elastic band [4, 5, 6, 7], by minimum-mode followers [8, 9, 10], and by string methods [11, 12]. The results are chained into adaptive kinetic Monte Carlo and accelerated dynamics [13, 14, 15, 16]. None of these is a

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single calculation. A long-timescale run in eOn [17, 18] spawns thousands of saddle searches, often on volunteer or distributed resources [19, 20]. It restarts them, and it passes geometries between Python drivers [21, 22], C++ engines and Fortran electronic structure codes [23]. The geometry file is therefore not an export format. It is the *restart state* of the algorithm.

That restart state has requirements that general-purpose formats were not designed for. A surface saddle search fixes the slab and lets an adatom move only along the surface normal. It therefore needs mobility per atom *and per axis*. Mode following and AKMC bookkeeping need atom identity that survives reordering. Dynamics-based steps need velocities whose unit is declared. eOn's natural time unit, $(\text{amu}\,\text{Å}^2/\text{eV})^{1/2} \approx 10.18\,\text{fs}$, differs from a femtosecond convention by a factor of about 103.6 in kinetic energy. A converged band restarted from rounded coordinates is no longer converged, so coordinates must round-trip in binary64 [24, 25, 26, 27]. Finally, two readers given the same bytes must agree, which needs a versioned grammar rather than a convention.

Existing formats meet some of these requirements, but none meets all of them. Extended XYZ carries per-axis masks only through an ASE-specific column. It writes coordinates at eight decimals by default, which moves positions by up to 10^{-9} Å on a round trip, and it has no version or velocity unit. H5MD [28, 29] records identity, velocities and box, but it has no per-axis constraint field and treats units as optional attributes. TNG [30, 31] and NetCDF trajectories [32] target compressed MD output, not optimiser restarts. ASE's own trajectory file is lossless within ASE, but it is defined by its Python implementation and cannot be read from the C++ and Fortran codes that do the work. Readers of the original CON format drift too. We found that the ASE copy of the eOn writer currently fails outright on per-axis constraints, and that a POSCAR round trip silently drops them. Toolkits such as chemfiles, MDAnalysis, Open Babel and IOData [33, 34, 35, 36] widen *access*, but they inherit whatever each format leaves undefined.

1.1. Contribution

We do not propose another container. We make the format eOn already uses into a *specification with a test suite* and give it a single implementation:

1. **CON 3.1**, a versioned grammar that specifies per-axis mask bits, a persistent atom identifier, velocities in a declared unit, and shortest round-trip binary64 output [37, 38].

2. **An executable conformance corpus**. Any reader, ours or not, can be checked against it. Agreement between readers is then something we test, not something we assume.
3. **One core, many bindings**. A Rust library behind a C ABI serves C++, Python, Julia [39] and Fortran. Per-language behaviour cannot drift, because there is only one parser. This follows the precedent set by chemfiles [33]. eOn 3.6.0 reads and writes through it.

We show each failure above as a reproducible case and show that CON 3.1 removes it. We then demonstrate bit-exact restart of a masked, asymmetric NEB across languages. Throughput comes last. We report it against the peer readers, with the cases where readcon is slower stated plainly [40, 41]. In line with FAIR software practice [42, 43, 44], the specification, corpus and implementation are versioned together [45].

Section 2 measures the five requirements on one frame. Section 3 states the grammar. Section 4 is the corpus and the extended XYZ disagreement. Section 5 is the reader and the store. Section 6 resumes one band through every route we completed. Section 7 reports threads, peers, MPI and the store. The release is readcon-core 0.17.1. The timings in Section 7 name the build that was clocked.

2. Requirements and the failure each one closes

Table 1 states five requirements. Each is a field a later step reads. The frame stores the geometry and its labels. It does not store the short history of a quasi-Newton update [46, 47] or of a FIRE step [48]. Section 6 measures that difference on one band.

R1 is a mask for each Cartesian axis of each atom. R2 is an integer stored on the row, so a later frame finds the same atom after the file groups rows by element. R3 is a time unit written in the file. R4 is coordinates, cell and periodic flags that read back as the values written. R5 is a grammar with a major and a minor, a rule for what a later minor adds, and a corpus that decides disputes.

2.1. One frame through each format

The frame is a Pt(100) slab with one adatom: the bottom atoms held on every axis, the adatom held in z, identifiers that are not the row order, and velocities in the natural unit. Section S7 gives the coordinates, the identifiers

Table 1: Requirements on a restart state, the step that reads each field, and the failure measured when the field is missing.

Requirement	Step that reads it	Failure measured here
R1, per-axis mask	Surface search, prefactor	Height constraint dropped; barrier $-70.2\,\mathrm{meV}$
R2, persistent id	Band and event table	Id lost on completed foreign round trips
R3, declared time unit	Dynamics	Kinetic energy scaled by 103.642696
R4, binary64 geometry, cell, periodicity	Resume of a search	10^{-9} to 10^{-6} Å, or periodicity rewritten
R5, versioned grammar	Any second reader	One file, four answers from five readers

and the grading rules. Each row of Table 2 writes that frame with one library and reads it with another. The libraries are ASE 3.29.0 [21], chemfiles 0.10.4 [33], MDAnalysis 2.10.0 [34, 49], and h5py 3.16.0 where the route needs it. POSCAR selective dynamics is the per-axis mask in the VASP ionic restart [50, 23].

The CON 3.1 row keeps every field. Extended XYZ written and read by ASE returns the mask, the identifiers and the time unit because both sides share ASE's column names and ASE's internal unit. The extended XYZ grammar does not define those names, so chemfiles drops the mask and MDAnalysis drops the mask and the identifiers. The ASE writer prints a fixed number of decimals and the coordinates move by up to 4.7×10^{-9} Å. chemfiles writing the same frame moves them by 4.9×10^{-6} Å and returns every axis periodic. MDAnalysis returns single-precision coordinates, a shift of 2.2×10^{-7} Å. The ASE trajectory keeps the mask and the coordinates exactly and drops the identifier. Its time unit is the library's unit, which the file does not name. The ASE POSCAR writer turns the held slab into a different mask and writes the adatom free. POSCAR has no atom identifier and no open axis. H5MD written by MDAnalysis declares the velocity unit, writes no mask and no identifier, stores single-precision coordinates, and marks the open z axis periodic [28]. An extra array carries a mask only when every reader agrees on its name and type. Section 6 attaches that array and shows which readers return it. The ASE eon reader rejects a CON

Table 2: One restart frame through each writer and reader. The frame is the Pt(100) slab with one adatom: the bottom layer is held on every axis, the adatom is held in z, ids are not the row order, and velocities are in the natural unit. R1 to R4 are measured on the round trip. "Kept" means the field came back. "Convention" means it came back only because writer and reader share one library's column name or internal unit, which the format does not define. "Fixed by format" means the format's own definition sets the unit without writing it in the file. R4 gives the largest coordinate change in Å when the coordinates are not bit-identical. Periodicity is the slab's xy-periodic, z-open flags after the read. R5 is the version rule in the format's published definition. "Error" rows failed before any field could be compared; the text gives the failure.

Format	Writer	Reader	R1 mask	R2 id	R3 unit	R4 coordinates	Periodicity	R5 version
CON 3.1	readcon-core	readcon-core	kept	kept	declared	exact	kept	major.minor
CON 3.1	readcon-core	ASE eon reader	error	error	error	error	error	major.minor
CON 3.1, coordinates only	readcon-core	ASE eon reader	full freeze	lost	lost	exact	none	major.minor
legacy CON	ASE eon writer	readcon-core	error	error	error	error	error	none
extended XYZ	ASE	ASE	convention	convention	convention	4.7×10^{-9}	kept	none
extended XYZ	ASE	chemfiles	lost	convention	lost	4.7×10^{-9}	not returned	none
extended XYZ	ASE	MDAnalysis	lost	lost	lost	2.2×10^{-7}	not returned	none
extended XYZ	chemfiles	ASE	lost	lost	lost	4.9×10^{-6}	all periodic	none
ASE trajectory	ASE	ASE	kept	lost	convention	exact	kept	code only
POSCAR	ASE	ASE	changed	lost	fixed by format	4.5×10^{-17}	all periodic	none
H5MD	MDAnalysis	MDAnalysis	lost	lost	declared	2.2×10^{-7}	all periodic	major.minor

frame that contains a velocity block. Given coordinates alone it freezes every masked atom on every axis, drops the identifiers, and returns no periodic axis. The ASE eon writer cannot express a per-axis constraint, so that route produces no file. Table S9 places the same checkpoint next to trajectory containers and structure archives.

3. The CON 3.1 specification

A CON file is a sequence of text frames. Listing 1 is one nine-atom frame. The survey frame of Table 2 uses the same columns.

Listing 1: A nine-atom Pt frame in CON 3.1. Line 2 declares the version, the units and the periodicity. The velocity block is abbreviated.

```
Pt(100) 2x2x2 with one Pt adatom
{"con_spec_minor":1,"con_spec_version":3,"pbc":[true,true,false],"sections
    ":["velocities"],"units":{"energy":"eV","length":"angstrom","mass":"
    amu","time":"(amu*angstrom^2/eV)^0.5"}}
5.5437171645025325e0 5.5437171645025325e0 1.396e1
9e1 9e1 9e1


1
9
1.95084e2
```

```
Pt
Coordinates of Component 1
1.3859292911256331e0 1.3859292911256331e0 6e0 7 0
4.1577878733769e0 1.3859292911256331e0 6e0 7 1
1.3859292911256331e0 4.1577878733769e0 6e0 7 2
4.1577878733769e0 4.1577878733769e0 6e0 7 3
0e0 0e0 7.96e0 0 4
2.7718585822512662e0 0e0 7.96e0 0 5
0e0 2.7718585822512662e0 7.96e0 0 6
2.7718585822512662e0 2.7718585822512662e0 7.96e0 0 7
1.3859292911256331e0 1.3859292911256331e0 9.76e0 4 8

Pt
Velocities of Component 1
1.4685859972256804e-3 6.275175116609037e-4 -1.8115929554652053e-3 7 0
3.748769356553213e-3 2.43327727576049e-3 -1.3238979663774354e-3 4 8
```

Line 1 is free text. Line 2 is one JSON object [51]. Lines 3 and 4 are the cell lengths and the angles in degrees. The type count, the per-type counts and the masses follow. Each coordinate row is x, y, z, the mask and the identifier. Lengths and angles rebuild the box with a along x and b in the xy plane,

$$\mathbf{a} = a\,\hat{\mathbf{x}}, \qquad \mathbf{b} = b\cos\gamma\,\hat{\mathbf{x}} + b\sin\gamma\,\hat{\mathbf{y}}, \tag{1}$$

after which $\mathbf{c}$ is fixed by α and β so the cell is right-handed [52]. When line 2 carries lattice vectors, those rows are the cell. A missing periodicity flag means periodic on every axis.

R1, the mask.. Column 4 is the integer of Table 3. Mask 0 is free, mask 4 fixes z only, and mask 7 fixes every axis. A float spelling, a sign, or a value outside 0 to 7 is an illegal flag. Under version 1 the column is 0 or 1, so masks 4 and 7 cannot be stored as written.

Table 3: Mask bits from version 2. Version 1 stores 0 (free) or 1 (fixed on x, y and z) in the same column.

Bit	Axis	Value
0	x	1
1	y	2
2	z	4

R2, the identity.. Column 5 is an integer from 0 through $2^{64} - 1$. Writers emit identifiers unique within a frame. A missing identifier column takes the zero-based row index.

R3, the time unit.. From minor 1, a velocity block declares `units.time`. The natural unit is spelled `(amu*angstrom^2/eV)^0.5`. Its value on the CODATA constants is 10.1805056712 fs [53]. Versions 1, 2 and 3.0 are read in the natural unit. An fs label on those older files is not trusted: producers of that era wrote natural-unit numbers under an fs label [18].

R4, the numbers.. Readers convert each finite decimal to binary64 with correct rounding [24, 25]. The default writer prints the shortest decimal that parses back to the same value [27, 38]. Six fixed decimals are an optional writer mode. gzip and zstd wrap the same bytes, and a reader recognises the wrapper by content [54, 55]. A conforming reader accepts the compressed file and returns the same frames.

R5, the version.. Table 4 lists the reserved keys of the JSON line. A reader accepts a later minor of a known major, keeps an unknown key on rewrite, skips an unknown section whose width is declared, and rejects an unknown major. H5MD uses the same split for its version pair [28]. Table 5 is the history of the grammar. Table 6 is the sections reserved at minor 1. The reader rules are Table S1. The grammar of one frame is the listing in Section S15. The prose specification wins where the listing and the prose disagree.

3.1. What the text costs

eOn has written CON since its first releases, so the campaigns already on disk are in this grammar [17]. The bytes do not depend on endianness [20]. A diff of a text checkpoint shows a changed mask or identifier. The JSON line takes a new key without a new grammar, and a reader keeps a key it does not know [51].

Table 7 is the byte cost of one 218-atom frame and of that frame repeated 10,000 times, raw and after gzip at level 6. The repeated file is the same text, so its gzip column is smaller than the gzip column of a trajectory whose frames differ. One CON frame is 41.99 bytes per atom, against 83.64 for extended XYZ written by ASE with the mask column, 314.39 for one H5MD frame written by MDAnalysis, and 95.40 for one H5MD frame written by h5py with binary64 positions, a mask dataset and an identifier dataset. At

Table 4: Reserved metadata keys of the JSON header line.

Key	Type	Key	Type
`con_spec_version`	int	`neb_bead`	int
`con_spec_minor`	int	`neb_band`	int
`section_arity`	object	`convergence_fmax`	number
`sections`	array	`convergence_energy`	number
`validate`	bool	`converged`	bool
`generator`	string	`fmax`	number
`units`	object	`bonds`	array
`pbc`	array	`storage_dtypes`	object
`lattice_vectors`	array	`chemfiles_atom_names`	array
`energy`	number	`chemfiles_atom_types`	array
`potential`	object	`chemfiles_atom_properties`	array or object
`frame_index`	int	`time`	number
`timestep`	number		

Table 5: CON versions. The date is the day the rule was fixed.

Version	Date	Change
1	eOn text	Column 5 present. Column 4 is 0 or 1. No JSON line.
2	2026-03-25	A JSON header line, identifiers, bit masks, declared sections.
3.0	2026-06-27	Length and energy units required. No minor.
3.1	2026-10-11	Minor 1. A velocity block declares the time unit.

10,000 copies the raw CON cost stays 41.99 bytes per atom. H5MD amortises its container and falls to 12.61 and 24.04 bytes per atom. gzip of the repeated CON file reaches 0.39 bytes per atom. H5MD is the record for a long trajectory [28]. CON is the record for the checkpoint, where one frame has to keep the mask, the identifier and the unit.

4. Conformance and drift

The corpus is the executable form of the grammar. It holds 10 legal files and 8 illegal files, split as in Table 8. A conforming reader accepts the first group and rejects the second. The corpus names five bindings, one per language of the C interface. The measured test of those languages on a

Table 6: Reserved sections at minor 1. Each data row carries its values, then the mask and the identifier.

Section	Values	Rule
Velocities	$v_x\, v_y\, v_z$	Time unit from minor 1
Forces	$f_x\, f_y\, f_z$	
Energies	one value	Name `energies`
Charges	one value	
Spins	one value	
Magmoms	$m_x\, m_y\, m_z$	
Displacements	$d_x\, d_y\, d_z$	`units.length`
Spreads	$s_x\, s_y\, s_z$	`units.length`, each at least 0

Table 7: Bytes per atom of the 218-atom frame, alone and repeated 10,000 times, raw and after gzip at level 6. CON 3.1 is the default readcon-core writer, extended XYZ the ASE writer with the mask column, H5MD (MDAnalysis) its writer in single precision, and H5MD (h5py) binary64 positions with mask and id datasets. The repeated file compresses better than any real trajectory would.

	One frame		10,000 frames	
Format	raw	gzip	raw	gzip
CON 3.1	41.99	6.07	41.99	0.39
extended XYZ (ASE)	83.64	6.65	83.64	1.13
H5MD (MDAnalysis)	314.39	9.04	12.61	0.12
H5MD (h5py)	95.40	9.66	24.04	0.16

resume is the 25-pair matrix in Section 6.

Table 9 applies R3 and R5 to seven headers of one velocity frame. Python and C follow the rule on each row. They reject a minor-1 velocity block with no time unit, an unknown major, a missing version and a mask of 9, and they keep an unknown key from a later minor. The ASE eon reader has no version rule. Given the coordinate block, it accepts all seven, including the illegal mask and the unknown major.

Extended XYZ has a published layout and parsers in several languages, so it is the text format to compare on R5. Table 10 gives five readers seven three-atom files. The readers are ASE 3.29.0, libAtoms 0.4.6 through its C path and its Python path, chemfiles 0.10.4, and MDAnalysis 2.10.0. On the reference frame the five readers return four outcomes: ASE and the libAtoms C path accept it, the libAtoms Python path changes the mask

Table 8: Conformance corpus. Counts are the committed manifest summary.

Category	Files	What a reader must do
Header and units	2	Accept a version-2 header and a version-3 units line.
Column-4 masks	5	Accept version-1 full freeze and version-2 per-axis flags.
Empty frame	3	Accept a type count of zero.
Illegal	8	Reject the file.

Table 9: Header variants of one velocity frame. Rule is what the specification requires. The ASE eon reader is given the coordinate block only, because it stops at a velocity section.

Variant	Rule	Python	C	ASE eon
3.1, natural unit declared	accept	accept	accept	accept
3.1, no time unit	reject	reject	reject	accept
3.0, labelled fs	accept (natural)	accept	accept	accept
3.2, unknown key	accept, keep	accept, kept	accept	accept
4.0	reject	reject	reject	accept
missing version	reject	reject	reject	accept
3.1, mask 9	reject	reject	reject	accept

booleans, chemfiles moves one coordinate by one unit in the last place and returns no mask, and MDAnalysis returns single precision with no mask and no identifiers. The correctly rounded binary64 value of the coordinate 2.7718585822512662 has bit pattern `40062cc4313f3fb9`; chemfiles returns `40062cc4313f3fb8` (Section S7). A Fortran exponent `6.0d0` is read by the libAtoms C path, returned as NaN by the libAtoms Python path, and rejected by the other three. A plain comment line is non-periodic to ASE, periodic to the libAtoms C path, and rejected by the libAtoms Python path. The readers follow their own documentation. The grammar does not decide these cases, and the formats share no corpus that would.

5. Reference implementation

readcon-core 0.17.1 is the reference reader and writer [56]. The concept record covers the line of releases [45]. Figure 1 is the layering. One Rust core parses and writes. One C interface is the waist every other caller uses. C++

Table 10: Five extended XYZ readers on seven three-atom files. Each file departs from the reference frame in one respect. ✓: positions correctly rounded, periodicity as written, mask and ids returned as written. E: the read failed. u: a coordinate one unit in the last place from the correctly rounded value. f: single-precision coordinates. n: a coordinate returned as NaN. p: periodicity differs from the file, or from the grammar's default when the key is absent. b: the mask came back with different booleans. m, i: the reader returns no mask, or no ids.

File	ASE	libAtoms C	libAtoms Py	chemfiles	MDAnalysis
reference frame	✓	✓	b	um	fmi
no pbc key	✓	✓	b	um	fmi
plain comment line	✓	p	E	u	f
Fortran exponent `6.0d0`	E	✓	nb	E	E
`True`/`False`	✓	✓	b	um	fmi
quoted value with `=`	✓	✓	b	um	fmi
`units` key	✓	✓	b	um	fmi

wraps it for eOn 3.6.0. Python exposes array views [57]. Julia calls the C interface [39]. Fortran 2008 calls the same interface. chemfiles took the same route for trajectories: one core and several language bindings [33]. Decimal fields go through one correctly rounded conversion on that shared path [27].

The caller owns a frame handle and frees it. Structure-of-arrays accessors return a pointer into that frame. Absent numeric metadata is a NaN sentinel and is not written back. The reader records the byte offset of each frame. The scan that finds those offsets is serial, because each start depends on the counts before it. Each frame is then parsed on its own, and the frames return in file order.

Listing 2: A round trip in Python. The default printer preserves binary64. Six decimals are a separate argument.

```
frames = readcon.read_con("slab.con")
readcon.write_con("rt.con", frames)
```

readcon-db is the campaign store for that text [58]. It writes each CON frame into a memory-mapped store, posts an xxHash3-128 of the text, and keeps secondary indexes on atom count, composition, energy, maximum force and section presence. A select touches the index and then the frames the index names. Identical texts share one hash entry, and each frame keeps its own stored text, so the stored blob bytes equal the raw file. gzip and zstd remain properties of the CON bytes, recognised by content before ingest. The MPI paths are a single writer, which inserts in global frame order, and

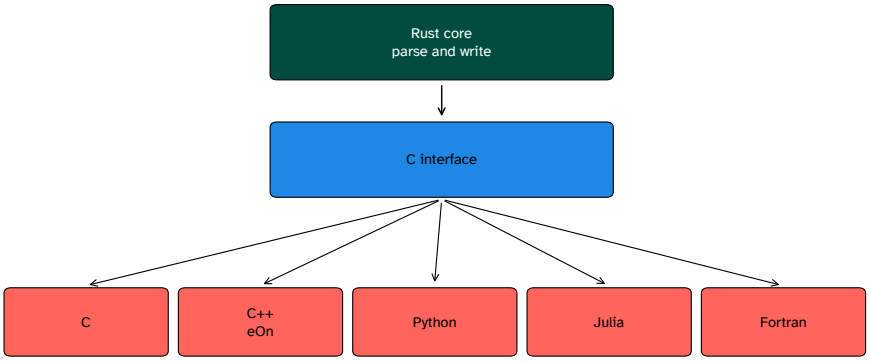


Figure 1: The Rust core parses and writes CON. The C interface is the waist. eOn 3.6.0 calls the C++ wrapper.

a sharded ingest. Section 7.3 measures both. The map starts at 2 GiB. The caller can raise that limit, and a write that fills the map grows it and retries. Section 7 names the versions that were clocked.

6. Case studies

Every route below uses one protocol. An EMT Pt(100) slab, three layers of 3×3, has its bottom layer held on every axis and one adatom held in z. The adatom hops between neighbouring hollows. Endpoints were relaxed to $0.002\,\mathrm{eV/Å}$. A climbing-image band of five movable images uses the improved tangent [5, 6] and fixed-step steepest descent on the projected force. Steepest descent stores no step history, so a difference after the resume comes from the file. The band is interrupted after 25 steps, written, read back, and resumed to $0.01\,\mathrm{eV/Å}$. Uninterrupted, it converges in 60 steps with barrier $0.644561366\,\mathrm{eV}$ and 56 free modes on the middle image. Table 11 is every completed route. Figure 2 is three of those bands. The 25 language pairs are Table S4.

The shortest CON round trip changes no coordinate. The barrier moves by $-5.68 \times 10^{-14}\,\mathrm{eV}$ and the images by at most $2.66 \times 10^{-15}\,\mathrm{Å}$. Each of the 25 language pairs also changes no coordinate, keeps 56 modes, and finishes in 60 steps at the same printed barrier. Their barrier shift is $-2.31 \times 10^{-14}\,\mathrm{eV}$ and their largest displacement is $3.55 \times 10^{-15}\,\mathrm{Å}$. The two floors are both at

Table 11: Climbing-image band of the EMT Pt(100) hop, stopped after 25 steepest-descent steps, written and read back, then resumed to 0.01 eV/Å. File is the largest coordinate change on that round trip. Modes are the free Cartesian degrees of freedom of the middle image (56 when the masks survive). $\Delta E^{\ddagger}$ and the displacement are against the uninterrupted band, whose barrier is 0.644561366 eV. [a]Not converged at 4000 steps. The 25 language pairs are Table S4.

Route	File (Å)	Modes	Steps	$\Delta E^{\ddagger}$ (eV)	Max displacement (Å)
uninterrupted	–	56	60	0	0
CON 3.1, shortest round trip	0	56	60	-5.68×10^{-14}	2.66×10^{-15}
CON 3.1, six decimals	4.90×10^{-7}	56	60	-4.14×10^{-7}	6.22×10^{-7}
CON 3.1 read by the ASE eon reader	0	54	4000[a]	-3.52×10^{-1}	5.43×10^{0}
same, periodicity restored by hand	0	54	127	-2.93×10^{-6}	4.21×10^{-2}
extended XYZ (ASE)	5.00×10^{-9}	56	60	1.09×10^{-9}	4.94×10^{-9}
ASE trajectory	0	56	60	-2.31×10^{-14}	3.55×10^{-15}
POSCAR (ASE)	4.90×10^{-17}	57	98	-7.02×10^{-2}	1.60×10^{-1}
legacy CON (ASE eon writer)	no file: a per-axis constraint is rejected				
ASE trajectory, id array	0	56	60	-2.31×10^{-14}	3.55×10^{-15}
H5MD MDAnalysis to MDAnalysis	4.60×10^{-7}	84	97	-7.07×10^{-2}	1.43×10^{-1}
H5MD MDAnalysis to h5py	4.60×10^{-7}	56	60	-1.98×10^{-7}	3.67×10^{-7}
H5MD h5py to h5py	0	56	60	-2.31×10^{-14}	3.55×10^{-15}
H5MD h5py to MDAnalysis	4.60×10^{-7}	84	97	-7.07×10^{-2}	1.43×10^{-1}

the level of rebuilding the calculator. The table keeps them separate. Six decimals move the file by 4.90×10^{-7} Å, the barrier by -4.14×10^{-7} eV, and the images by 6.22×10^{-7} Å, and the band still converges in 60 steps with 56 modes. Extended XYZ moves the file by 5.00×10^{-9} Å and the barrier by $+1.09 \times 10^{-9}$ eV, with the same step count and the same mode count. The ASE trajectory matches the language-pair floor. An extra identifier array on that trajectory does not survive the round trip, and the mode count stays 56 because the mask does.

POSCAR moves the file by 4.90×10^{-17} Å and drops the height constraint, so the middle image has 57 free modes. The band converges in 98 steps, 0.160 Å away, with a barrier 7.02×10^{-2} eV lower. That is the 70.2 meV in the abstract. The ASE eon writer produces no file, because a per-axis constraint is not a constraint that writer emits. The ASE eon reader keeps 54 modes, runs to the 4000-step cap without converging, and finishes 0.352 eV and 5.43 Å away. Restoring periodicity by hand leaves the mask collapse in place: 54 modes, 127 steps, a barrier shift of -2.93×10^{-6} eV, and a displacement of 4.21×10^{-2} Å.

The H5MD block in the table is a second measurement, distinct from the survey row. The writer attaches a Boolean $N \times 3$ mask array, an identifier array, the velocity unit Å ps^{-1}, and the boundary periodic, periodic, none.

MDAnalysis reading that file, whether it wrote the file or h5py did, returns 84 free modes and a barrier 7.07×10^{-2} eV lower, with a file shift of 4.60×10^{-7} Å. The mode count is the loss of the mask and the identifier. h5py reading the MDAnalysis file keeps 56 modes, finishes in 60 steps, and moves the barrier by -1.98×10^{-7} eV. h5py reading its own file is bit-exact and lands on the language-pair floor. The coordinate gap between the two files is the single-precision write.

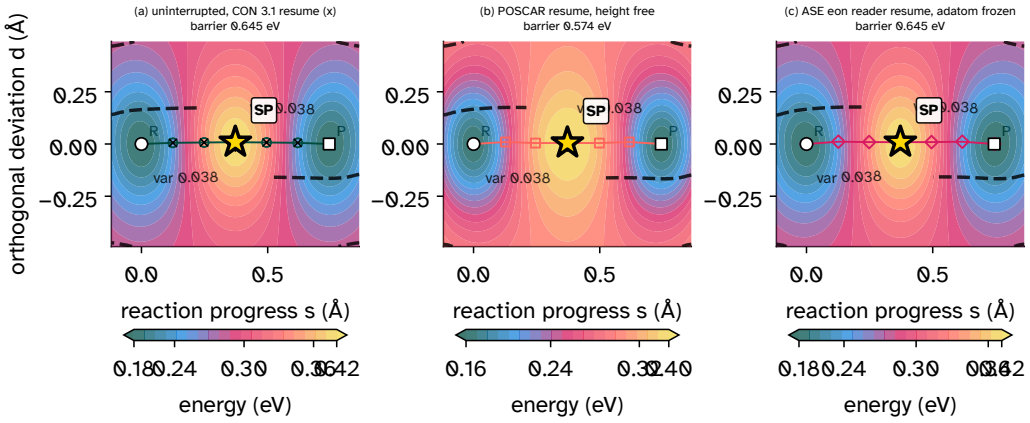


Figure 2: Resumed climbing-image bands of the EMT Pt(100) hop on the (s, d) plane [59]. Each panel is the energy surface of its own band. (a) Uninterrupted, with the CON 3.1 resume as crosses. (b) POSCAR, height constraint dropped. (c) ASE eon reader with periodicity restored by hand. The routes differ in barrier and in free modes, as Table 11 records.

6.1. The mask and the prefactor

The symmetric hop can sit on the straight line, so a lost lateral constraint need not move the barrier by much. A second band, seven images on the smaller slab, was converged with FIRE to 0.01 eV/Å [48]. ASE 3.29.0 has no eOn Morse calculator in this run, so the relaxation, the band and the Hessian use EMT. With the height held, the barrier is 0.36659812266626624 eV on a path of 2.9004016648060382 Å. The middle image has 56 free modes and one imaginary mode, −4.076276454192005 meV, with 0.9872893953755115 of its weight on the adatom's lateral motion. The Vineyard prefactor of that Hessian is $3.2438730218892426 \times 10^{10}$ Hz [2]. FIRE took 26 steps and then 16 climbing steps.

The ASE eon reader turns the height-held column into a full freeze, so the adatom is fixed on every axis. The default interpolation then stops because the constraints change the path, and the band was built with the constraints ignored during interpolation. FIRE took 26 steps and then 14 climbing steps, reached 0.007095 eV/Å, and left the adatom on the straight interpolation. The relative peak is 0.3665836803750313 eV on a path of 2.9595547559012947 Å. The middle image has 54 free modes and no imaginary mode, so the Vineyard ratio is undefined. Figure 3 is the two bands. The barriers differ in the fifth digit. The prefactor exists for one reading and does not exist for the other.

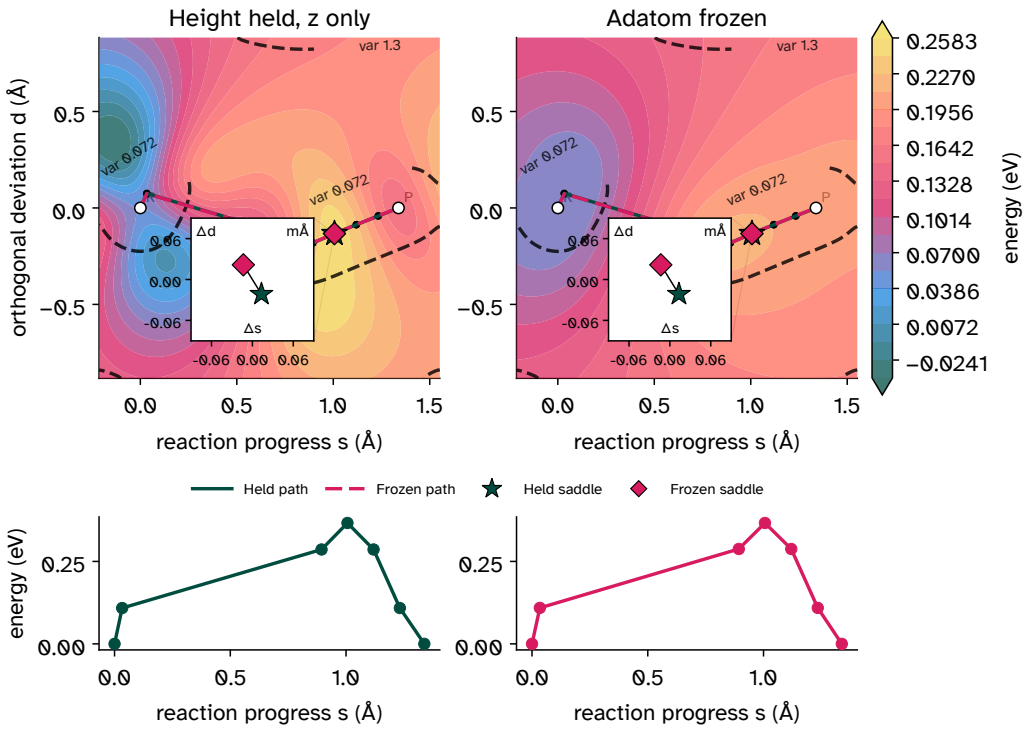


Figure 3: EMT climbing-image bands for a Pt adatom hop on Pt(100), each mask on its own (s, d) map [59]. Height held: 0.366598 eV, 56 free modes, one imaginary mode. ASE eon reading: 0.366584 eV, 54 free modes, no imaginary mode.

6.2. eOn resuming a Morse band

An eOn 3.6.0 client ran a Morse Pt potential on a 28-atom slab, bottom atoms at mask 7, five movable images, steepest descent [18]. The band

section selects the optimiser. This run sets steepest descent there. When the total number of optimiser steps matches, the shortest round trip reproduces the uninterrupted barrier, 0.63493 eV, with barrier difference 0 and a largest minimum-image displacement of $8.881784197001252 \times 10^{-15}$ Å. Six decimals give barrier 0.634931 eV and a largest minimum-image displacement of $4.918050526825368 \times 10^{-7}$ Å. Figure 4 is that comparison. Section S1 gives the per-image displacement.

A control file left the default band optimiser in place. Both printers of that control finish at 0.670723 eV. The uninterrupted barrier in the same file is 0.63361 eV, so the difference is 0.037113 eV. The two printers agree with each other. They do not agree with the uninterrupted barrier, because the checkpoint does not carry the optimiser history.

A dimer started from the round-trip geometry stops for the same reason as the uninterrupted dimer, too many iterations. Both report saddle energy −112.618941 eV and curvature −10.613681. Six decimals give −112.618944 eV and −10.613677, an energy shift of -3×10^{-6} eV. The comparison is of the stopped search.

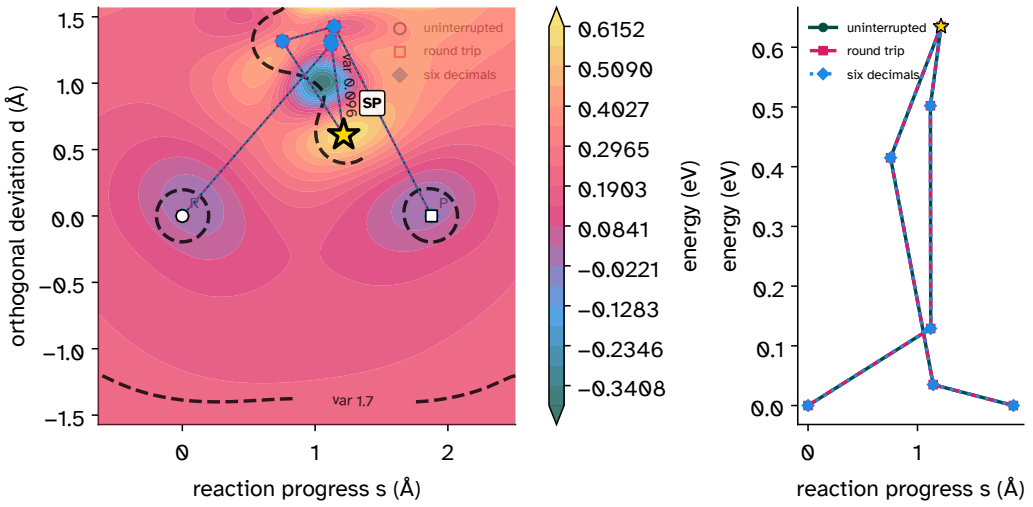


Figure 4: Final images of the Morse band on the (s, d) plane [59]. With the step count matched, the shortest round trip returns the uninterrupted barrier, 0.63493 eV. The colour bar is the fitted surface. The image energies are the energy column of the uninterrupted band, and the highest image is 0.634930 eV.

6.3. The velocity unit and identity

On the 28-atom slab the kinetic energy is 0.8878595057627452 eV in the natural unit and 92.0201533142073 eV when those numbers are read as Å/fs. The ratio is 103.64269652680504. One velocity Verlet step of 1 fs then moves the atoms by an RMSD of 0.09556596694568728 Å, maximum 0.31544697368347396 Å [60]. The same factor appears on a separate 10,001-atom slab at 300 K: the natural unit is 10.1805056712 fs, the kinetic energy is 390.085819 eV, and the misread value is 40429.545793 eV, a factor 103.642696. Section S4 plots the step.

The public HCN and HNC endpoints carry the unit and the identity in one file [61, 62]. readcon-core on frame 2 returns masks `000 000 100`, identifiers `2 0 1`, and the declared natural unit. The ASE eon reader stops on the velocity block. Given coordinates only, it returns the three atoms as free, free, and fixed on every axis, and it returns no identifiers.

7. Performance

Timings below are medians. Where an interval is reported, it is a bias-corrected and accelerated 95 percent interval of the median from 9999 resamples [63]. The host, the compiler and the thread pin are part of the result [41, 40]. Parallel runs are compared at fixed work or fixed work per rank. The release is readcon-core 0.17.1. The clocked builds are older, and this section names them.

Thread timings and peer campaign A used an Intel(R) Xeon(R) Processor, family 6, model 207, 8 hardware threads, rustc 1.88.0, g++ 13.3.0 (Ubuntu 13.3.0-6ubuntu2 24.04.1) and Python 3.12.3. TODO(verify): commercial CPU SKU, clock frequency, RAM and operating-system release for that campaign. Peer campaign B used an Intel(R) Xeon(R) Processor, 8 hardware threads, rustc 1.99.0 and Python 3.12.15. TODO(verify): commercial CPU SKU, family and model, clock frequency, RAM, operating-system release and C compiler for campaign B. The MPI campaign used an Intel(R) Xeon(R) Processor, family 6, model 207, stepping 2, 8 cores, 1 thread per core, 1 socket, 49,342,904 kB of memory, Linux 6.12.94+, rustc 1.99.0, GCC 13.3.0 (Ubuntu 13.3.0-6ubuntu2 24.04.1) and MPICH 4.2.3. TODO(verify): commercial CPU SKU, clock frequency and operating-system distribution for the MPI host. The kernel string is recorded. The store campaign used an Intel(R) Xeon(R) Processor, 8 CPUs, 47.07 GiB, Python 3.12.3 and rustc

1.88.0. TODO(verify): commercial CPU SKU, family and model, clock frequency and operating-system release for the store host. None of the logs records a clock frequency, so none is stated.

7.1. Threads

The file is 10,000 copies of one 218-atom frame, 167,710,000 bytes, read by readcon-core 0.17.0 built with rustc 1.88.0. OpenMP and BLAS are pinned to one thread outside the pool under test. Table 12 is the warm-cache read. The native Rust pool finishes in 0.125501 s, 1336 MB/s, 79681 frames/s, which is 12.6 μs per frame. The C++ wrapper eOn calls finishes in 0.13315 s. The Rust-to-C++ ratio of those medians is 0.904, with interval $[0.815, 0.993]$. At one thread the two medians are 0.483641 s and 0.484879 s, ratio 0.991. Building a Python object per frame takes 1.234571 s. One EMT force call on this host has median 5.2623×10^{-5} s, so the native read of the whole file is about 2385 such calls. The in-process Morse timing from the band study is 3.036×10^{-4} s per call, and the same file is about 413 of those calls.

Table 12: Warm read of 10,000 frames, 167,710,000 bytes, readcon-core 0.17.0. Native rows are 15 repeats. The Python row is 9 repeats. The interval is a bias-corrected and accelerated 95 percent interval of the median.

Reader	Median (s)	MB/s	Interval (s)
Rust, automatic pool	0.125501	1336	[0.115354, 0.137084]
C++ wrapper, automatic pool	0.13315	1260	[0.120279, 0.153288]
Rust, 1 thread	0.483641	346.8	[0.473807, 0.486509]
C++ wrapper, 1 thread	0.484879	345.9	[0.480843, 0.537422]
Python, full objects	1.234571	135.8	[1.216353, 1.303821]

A separate five-repeat sweep on the same file sets the pool size. Rust goes from 0.50026 s at one thread to 0.11503 s at eight, a speedup of 4.35. C++ goes from 0.50177 s to 0.12793 s, a speedup of 3.92. Python full objects go from 1.2755 s to 1.1070 s, a speedup of 1.15. A second Python sweep, thirty interleaved pairs on the same file, is a different campaign. Its eight-thread speedup is 1.284, with interval $[1.263, 1.313]$. Section S14 records that sweep. Hashes of the returned fields at 1, 2, 3, 4 and 8 threads are identical. Figure 5 is the automatic-pool rates. Figure 6 places the scaling-block rates under the memory-copy rate. The serial fraction drawn there is 0.267.

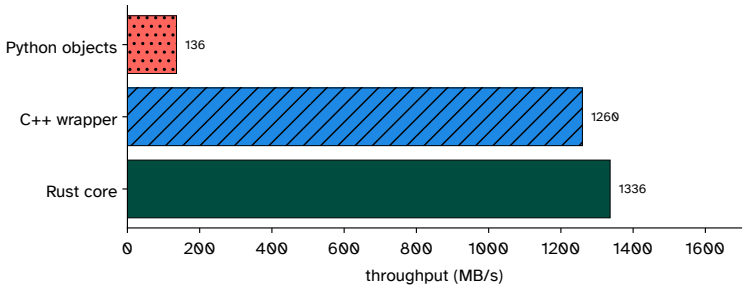


Figure 5: Warm-cache rate for the automatic Rust pool, the C++ wrapper and the Python full-object read, from the medians in Table 12.

7.2. Peers

Table 13 is readcon-core through its Python binding at one worker, divided by the peer, so a ratio below 1 means readcon-core finished sooner. Campaign A, readcon-core 0.17.0 and rustc 1.88.0, thirty interleaved pairs, compares the ASE eon reader, chemfiles XYZ, h5py and MDAnalysis H5MD. On 10,000 frames the ratios are 0.337 against ASE eon (readcon-core 1.269 s, ASE 3.745 s), 2.15 against chemfiles XYZ, 67.1 against h5py (peer 0.01899 s), and 0.207 against MDAnalysis H5MD (peer 6.179 s). h5py returns one numeric array. The CON reader returns the mask, the identifier, the cell and the periodic flags. Campaign B, rustc 1.99.0 and Python 3.12.15, adds ASE extended XYZ and the ASE trajectory. The ASE eon ratio on the same 10,000-frame file is 0.348, against 0.337 in campaign A. Absolute readcon-core times differ between the campaigns, near 2.0 s in B and 1.27 s in A, so only the reproduced ratio is compared. On that long file the ASE trajectory is level with readcon-core (ratio 1.02). On one slab frame the ASE trajectory is faster (ratio 3.19). Section S9 extends the table to the 1-frame and 100-frame files. On one frame the open dominates, and h5py and chemfiles lose the lead they have on the long file.

7.3. MPI

The MPI campaign used readcon-core 0.17.0 and readcon-db 0.1.8 on the eight-core host named above. Ranks 16 and 32 were not run. MPICH 4.2.3 used the ch4:ofi device, bound to cores, with UCX and libfabric on TCP. Each rank is one thread. Each cell is five repeats. Cold repeats drop the page cache. Warm repeats follow an untimed warmup. The median is the

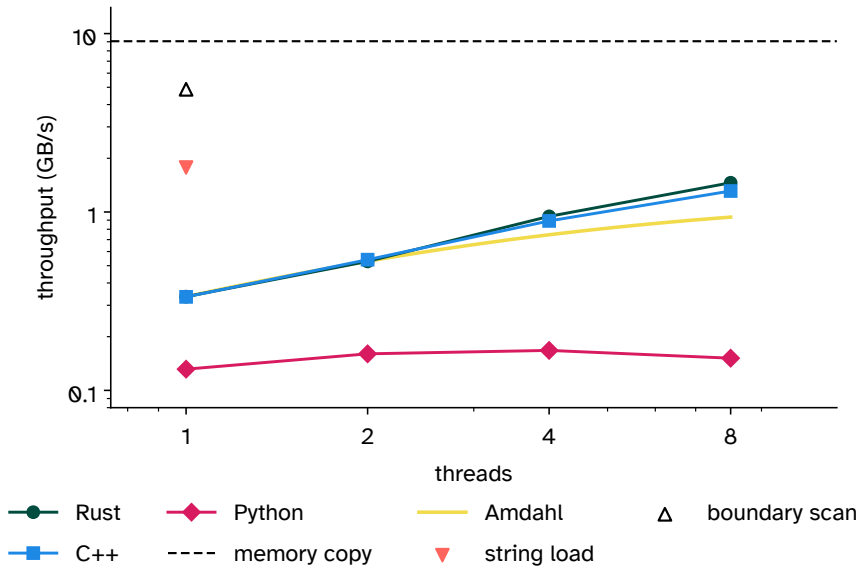


Figure 6: Scaling-block rates for Rust, C++ and Python on readcon-core 0.17.0. The dashed line is the hot-cache copy, 9.06 GB/s. The curve is Amdahl's law anchored at the one-thread Rust rate of the scaling block. Its serial fraction is 0.267.

third of the five sorted times. Every one of the 480 repeats matched the serial bytes.

Strong scaling holds the file fixed. The CON file is 1,583,691 copies of a small frame, 3,167,382 frames, 1,073,742,498 bytes. The database file is 30,965 frames of 1,000 atoms, ingested four times, with four shards on the sharded path. Table 14 is the warm medians. Shared read goes from 9.088 s to 1.504 s, a ratio 6.04. Scatter goes from 9.223 s to 1.836 s, ratio 5.02. MPI-IO write goes from 8.369 s to 1.416 s, ratio 5.91. Gather write goes from 9.490 s to 2.152 s, ratio 4.41. One-writer ingest goes from 31.699 s to 16.810 s, ratio 1.89. Four-shard ingest goes from 32.549 s to 18.805 s, ratio 1.73. Cold shared read goes from 9.904 s to 2.113 s. Figure 7 is the warm speedup. The full rank list, cold and warm, is in Section S12.

Weak scaling gives every rank the same frame count as the one-rank run. readcon-core holds 400,000 frames per rank. readcon-db ingests 7,700 frames of 1,000 atoms per rank, four copies. Table 16 is the warm medians, and the ratio is the eight-rank time over the one-rank time. Shared read grows by 1.36, scatter by 1.61, MPI-IO write by 1.33, and gather write by 1.78. One-writer ingest grows by 3.77, from 7.841 s to 29.594 s. Four-shard ingest grows

Table 13: Paired ratios of the one-worker Python read to each peer, readcon-core time over peer time, so a value below 1 means readcon-core finished sooner. Each cell is 30 interleaved pairs; the interval is a bias-corrected and accelerated 95 percent interval of the median. The fields column says what the peer returns. The two ASE rows beyond the eon reader come from a second campaign on a shared host, in which the ASE eon ratio on the 10000-frame file reproduced at 0.348 against 0.337.

Payload	Peer	Fields	Ratio	Interval
10000 frames	ASE eon	same CON text	0.337	[0.335, 0.345]
10000 frames	chemfiles XYZ	coordinates	2.15	[2.14, 2.17]
10000 frames	h5py	one binary array	67.1	[66.5, 67.3]
10000 frames	MDAnalysis H5MD	coordinates	0.207	[0.204, 0.213]
Pt slab	ASE eon	same CON text	0.233	[0.231, 0.241]
Pt slab	chemfiles XYZ	coordinates	1.14	[1.13, 1.16]
Pt slab	h5py	one binary array	6.05	[5.91, 6.27]
Pt slab	MDAnalysis H5MD	coordinates	0.394	[0.387, 0.404]
10000 frames	ASE extended XYZ	ASE atoms	0.392	[0.338, 0.42]
10000 frames	ASE trajectory	ASE atoms	1.02	[0.948, 1.08]
Pt slab	ASE extended XYZ	ASE atoms	0.184	[0.18, 0.192]
Pt slab	ASE trajectory	ASE atoms	3.19	[3.08, 3.28]

by 4.68, from 7.465 s to 34.899 s. Every rank writes one virtio ext4 device, and the byte count grows with the rank count. Table 15 is the aggregate rate on that device. Shared read and MPI-IO write still gain rate from one rank to eight. Both ingest paths level off, and at eight ranks the sharded path is the slower of the two on the same bytes. Figure 8 is that growth.

7.4. The store

The store measurements used readcon-db 0.1.8 and readcon-core 0.16.2 [64] on the store host named above. The archived store release is 0.2.0 [58]. Threads are pinned to 1. Five repeats are interleaved. A cold query asks the kernel to drop the file from the cache and then runs in a fresh process. The run did not drop the page cache. A warm query is the next query in that process. The timed corpora have 1,000 and 10,000 frames, and every frame has its own xxHash3-128. The frames are ASE effective-medium geometries stored as CON text. Section S10 records the repeats and the figures.

At 10,000 frames the ingest medians are 0.311306 s for readcon-db (32,123 frames/s, 35.09 MB/s), 48.411 s for ASE's database committing every write (206.6 frames/s), and 5.985 s for one ASE connection that still commits every 5,000 rows (1,671 frames/s). The paired ratios are 0.006800 against the

Table 14: Strong scaling, warm page cache. Each cell is the median of five repeats. t_1/t_8 is the one-rank median divided by the eight-rank median. The file is 1,073,742,498 bytes. readcon-core holds 3,167,382 frames. readcon-db ingests four copies of a 1,074,071,389-byte trajectory (30,965 frames of 1,000 atoms), batch 4096. One thread per rank. Every output matches the serial bytes.

Path	1 rank (s)	8 ranks (s)	min–max at 8	t_1/t_8
read, shared file	9.088	1.504	1.416–1.570	6.04
read, scatter	9.223	1.836	1.810–1.875	5.02
write, MPI-IO	8.369	1.416	1.366–1.491	5.91
write, gather	9.490	2.152	2.102–2.262	4.41
ingest, one writer	31.699	16.810	16.663–17.330	1.89
ingest, four shards	32.549	18.805	18.644–18.879	1.73

Table 15: Warm weak scaling on one virtio ext4 device. The aggregate rate is `bytes_total` divided by the median, and a megabyte is 10^6 bytes. The ratio is the median at that rank count divided by the one-rank median of the same path. Shared read and MPI-IO write are the endpoints. Both ingest paths are the full rank list, which is where the rate flattens.

Path	Ranks	Time (s)	Bytes	MB/s	$t(R)/t(1)$
shared read	1	1.161	135,600,000	116.8	1.00
shared read	8	1.5815	1,084,800,000	685.9	1.36
MPI-IO write	1	1.0369	135,600,000	130.8	1.00
MPI-IO write	8	1.3821	1,084,800,000	784.9	1.33
one-writer ingest	1	7.841	1,068,323,900	136.2	1.00
one-writer ingest	8	29.594	8,546,831,576	288.8	3.77
four-shard ingest	1	7.465	1,068,323,900	143.1	1.00
four-shard ingest	2	10.856	2,136,675,340	196.8	1.45
four-shard ingest	4	18.375	4,273,391,820	232.6	2.46
four-shard ingest	8	34.899	8,546,831,576	244.9	4.68

per-write commits and 0.05283 against the single connection. At 10,000 frames the readcon-db query medians are 8.14 ms, 10.1 ms and 9.57 ms cold for composition, energy and maximum force, and 0.176 ms, 0.211 ms and 0.208 ms warm. ASE's database takes about 0.2 s on the same queries, cold or warm. A directory of the same text takes about 0.14 s warm. readcon-db occupies 2.503 times the raw CON text: the text plus the indexes. The map used for these tables starts at 2 GiB, so a one-million-frame open exceeds it and stays outside the tables.

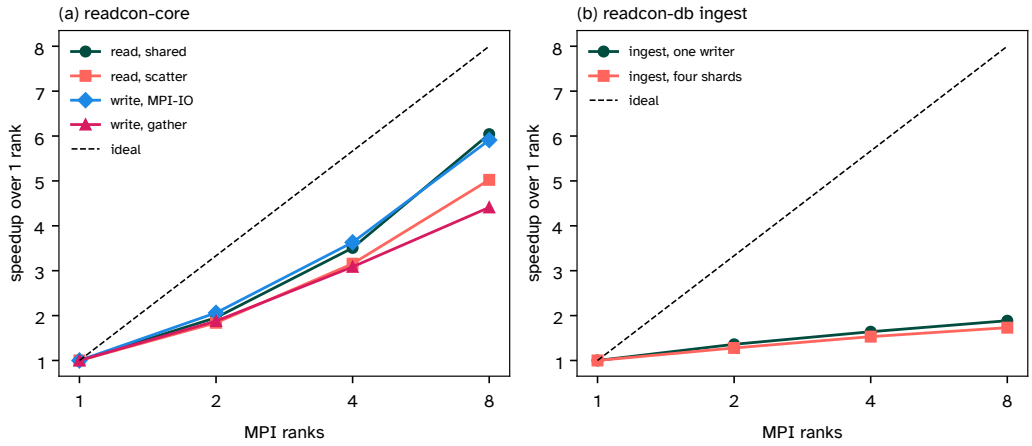


Figure 7: Strong-scaling speedup from the warm medians in Table 14. The dashed line is ideal speedup. Ranks 1, 2, 4 and 8 on the 8-core host. Every output matched the serial bytes.

8. Conclusions

The geometry file in a rare-event campaign is the restart state of the search. The resume reads a per-axis mask, a persistent identifier, a declared time unit, binary64 coordinates with their periodic flags, and a version two readers agree on. On one masked slab, every common format tested here drops at least one of those fields. The ASE POSCAR resume drops the height constraint and finishes 70.2 meV lower. Five extended XYZ readers return four answers for one file. CON 3.1 states the grammar, the corpus decides the listed files, and one parser serves the five languages. Twenty-five language pairs resume the climbing-image band at the uninterrupted barrier of 0.644561366 eV. With the height held, the Vineyard prefactor of the EMT hop is $3.2438730218892426 \times 10^{10}$ Hz. The ASE eon reading of the same band has no imaginary mode, so that ratio is undefined. A warm read of 10,000 frames of 218 atoms takes 12.6 μs per frame on the 0.17.0 build named in Section 7. On that file chemfiles is 2.15 times as fast and h5py is 67.1 times as fast. Shared MPI reads speed up by 6.04 from one rank to eight on this host. Under weak scaling the four-shard ingest grows by 4.68 because every rank writes the same device. The release is readcon-core 0.17.1.

Table 16: Weak scaling, warm page cache. Each rank holds the same frame count as the one-rank run. t_8/t_1 is the eight-rank median divided by the one-rank median. readcon-core holds 400,000 frames per rank (135,600,000 bytes at one rank). readcon-db ingests 7,700 frames of 1,000 atoms per rank, four copies (30,800 frames per rank). Medians of five repeats. Every output matches the serial bytes.

Path	1 rank (s)	8 ranks (s)	min–max at 8	t_8/t_1
read, shared file	1.161	1.581	1.573–1.792	1.36
read, scatter	1.218	1.962	1.949–2.264	1.61
write, MPI-IO	1.037	1.382	1.365–1.446	1.33
write, gather	1.211	2.161	2.097–2.296	1.78
ingest, one writer	7.841	29.594	28.943–30.364	3.77
ingest, four shards	7.465	34.899	34.807–36.533	4.68

Data availability

readcon-core 0.17.1 is https://doi.org/10.5281/zenodo.23405710 [56]. The concept record is https://doi.org/10.5281/zenodo.18753567 [45]. readcon-core 0.16.2, the reader used for the store timings, is https://doi.org/10.5281/zenodo.23264985 [64]. The companion store is archived at https://doi.org/10.5281/zenodo.23652386 [58]. The specification, the grammar, the corpus and the campaign results are the supplementary material.

CRediT authorship contribution statement

Rohit Goswami: Conceptualization, Methodology, Software, Visualization, Writing – original draft, Writing – review and editing.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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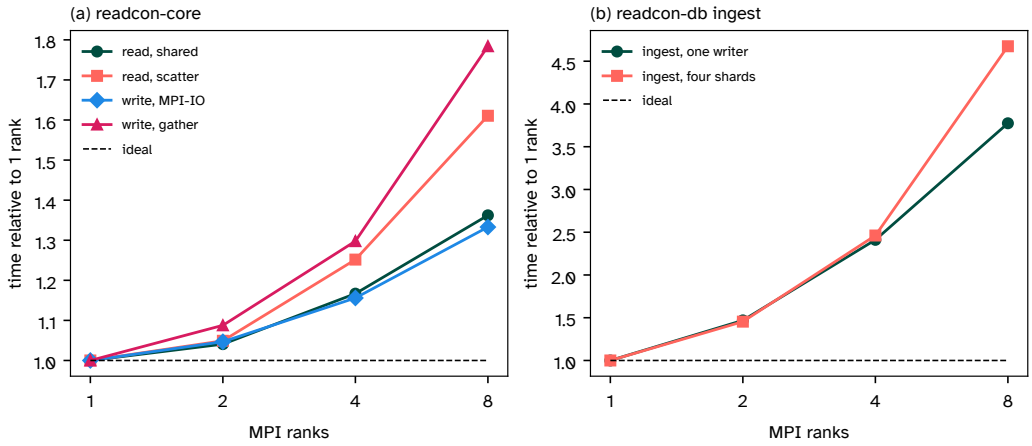


Figure 8: Weak scaling from the warm medians in Table 16. The dashed line is a ratio of 1, the same time as one rank. The ingest ratios lie above that line.

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